

Summary of Revisions

Interpretable Multivariate Conformal Prediction with Fast Transductive Standardization

JMLR-25-3153-1

This summary compares the integrated revision with the original arXiv v1 submission. It distinguishes the supplied author’s edits from the newly integrated experiments; it does not certify new theoretical results.

- **Abstract and Section 1: motivation and related work.** The central motivation and contribution statement are retained; no substantive rewrite is claimed. **Author check:** Correct the Point CHR/CQHR distinction and add shape-template related work. These changes are necessary for accurate positioning of the new comparisons.
- **Section 2: setup and motivating constructions.** The supplied revision renames and reorganizes the preliminaries, clarifies joint exchangeability and simultaneous versus marginal coverage, strengthens vector/scalar distinctions, discusses ties, and introduces a common parameterized population-oracle formulation. This makes the statistical target and the role of standardization more explicit. **Author check:** Reconcile capped-score ties and shift claims with the actual implementation; define or weaken the remaining “uniformly tight” wording.
- **Section 3: methodology.** The supplied edit uses a common threshold-indexed prediction-set map, separates observed and oracle scaling statistics, rewrites the global/local construction in that notation, and corrects the exceptional empty-cell sign. This improves traceability of the conservative approximations. **Author check:** Review the new working-hypothesis conditional-coverage claim, the search-lemma endpoint inequality, and the piecewise coverage proof.
- **Section 4: numerical experiments.** New main-text material includes dependent heterogeneous Gaussian noise, coordinate-length bar plots, partial heteroskedasticity, native CQHR, and real-data wall-clock comparisons. New 200-split real-data reruns add all coordinate lengths and direct backward-search/fallback counts. Fresh-draw protocols and CQR model settings are explicit. The original energy example remains in the body. These changes answer requests for coordinate detail, broader baselines, and computational evidence; the Point CHR infinite-volume case is now explained by its allocation.
- **Section 5: discussion.** The existing discussion is retained without alteration. **Author check:** Connect it to the new contamination and CQR evidence while retaining the distinction between empirical robustness and proved validity.
- **Appendix A: algorithms and complexity.** The supplied algorithmic material is retained. No faster worst-case bound is claimed. The new Section 4 diagnostics measure actual search branches and their observed costs, rather than inferring branch frequency from overall runtime.
- **Appendix B: proofs.** The supplied proof section is preserved, including its updated notation. No new moment-relaxation or tie-handling theorem is claimed. **Author check:** Check mathematical consistency with the revised main text; this integration did not perform a complete proof-equivalence audit.
- **Appendix C: new sensitivity and stress tests.** The added analyses cover uncertainty, dependence, heterogeneity, target level, small samples, heavy tails, contamination, CQR base widths and constant shifts, and a disclosed low-dimensional shape-template implementation. New real-data marginal-coverage and search-regime stress studies complete the diagnostic evidence. These analyses identify where the empirical benefits persist and where efficiency or theoretical scope is limited.
- **Appendix D: retained original experiments.** The original Gaussian figures, distribution tables, oracle/local-union and dimension-scaling studies, heavy-tail figures, and original real-data table are retained here, rather than deleted. Their historical pool-resampling protocol is distinguished from the new independent draws. This preserves earlier evidence while keeping reviewer-focused comparisons in the body.
- **Figures, bibliography, and remaining completion.** All original experimental assets and labels are retained; 19 new PDF figures and the Tumu et al. reference are added. Figures 1 and 2 now show illustrative marginal coverage rates and residual-boundary labels, respectively, to clarify their interpretation. **Author check:** Finish notation consistency and manuscript-wide color marking. The separate reviewer audit lists the remaining items.